

Experiment 3: Working with Amazon DynamoDB

Amazon DynamoDB is a **fully managed NoSQL database service provided by** Amazon Web Services. It is used to store and retrieve large amounts of data with **high performance and scalability**.

DynamoDB stores data in **tables**.

Each table contains **items (records)** and **attributes (fields)**.

Every table must have a **primary key** to uniquely identify each item.

It provides **fast data access with low latency**.

It automatically handles **scaling, backup, and maintenance**.

1. Login to AWS

Open **AWS Management Console**.

Login with your AWS account.

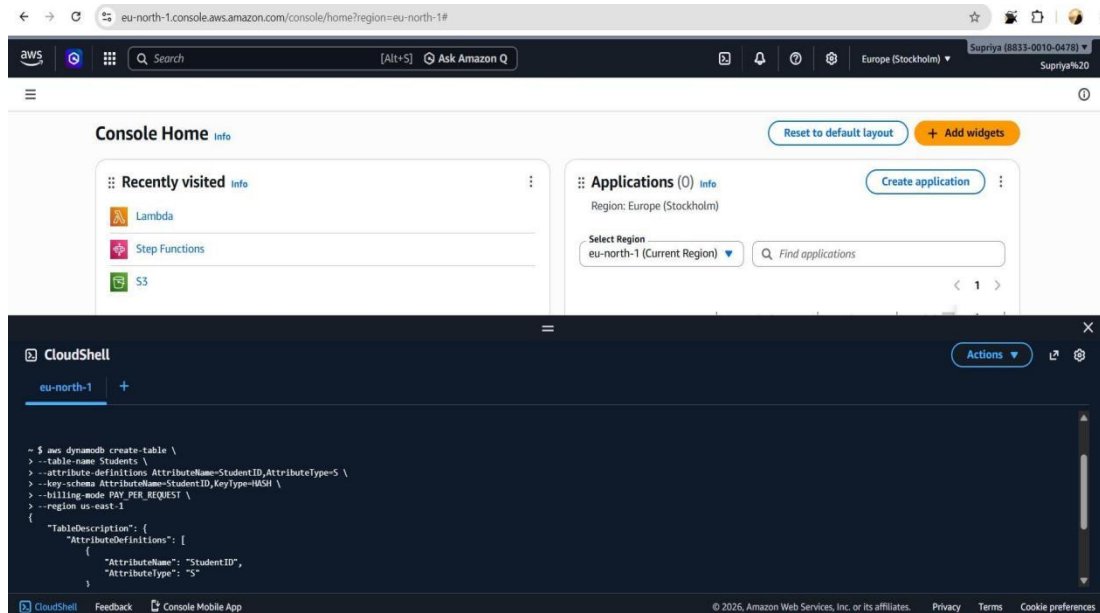
Search for **CloudShell** in the top search bar.

Open **CloudShell Terminal**.

2. Create a DynamoDB Table

Run the following command in CloudShell.

```
aws dynamodb create-table \  
--table-name Students \  
--attribute-definitions AttributeName=StudentID,AttributeType=S \  
--key-schema AttributeName=StudentID,KeyType=HASH \  
--billing-mode PAY_PER_REQUEST \  
--region us-east-1
```



Output :-

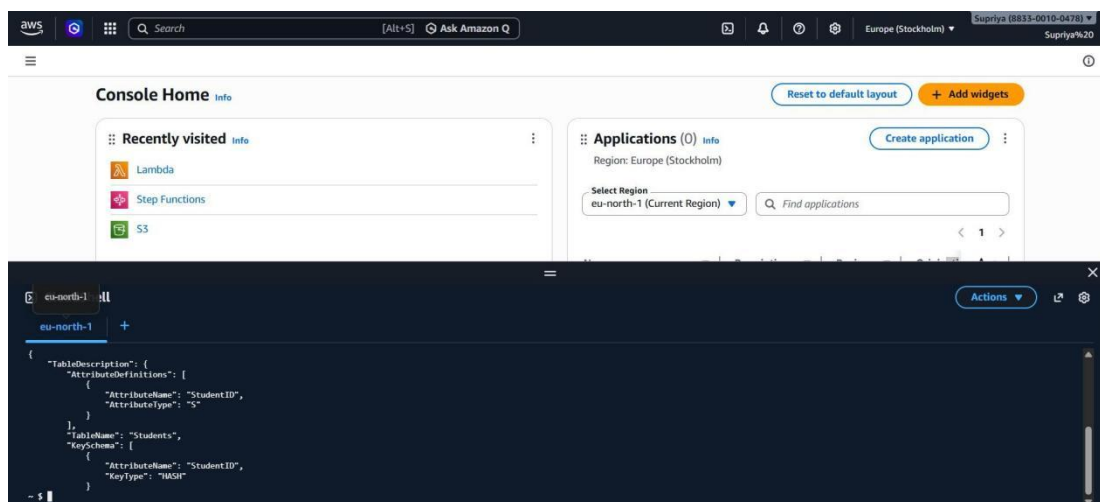


Table Does Not Exist

The **Students** table may not be created.

Check using:

aws dynamodb list-tables --region us-east-1

3. Insert Data into the Table

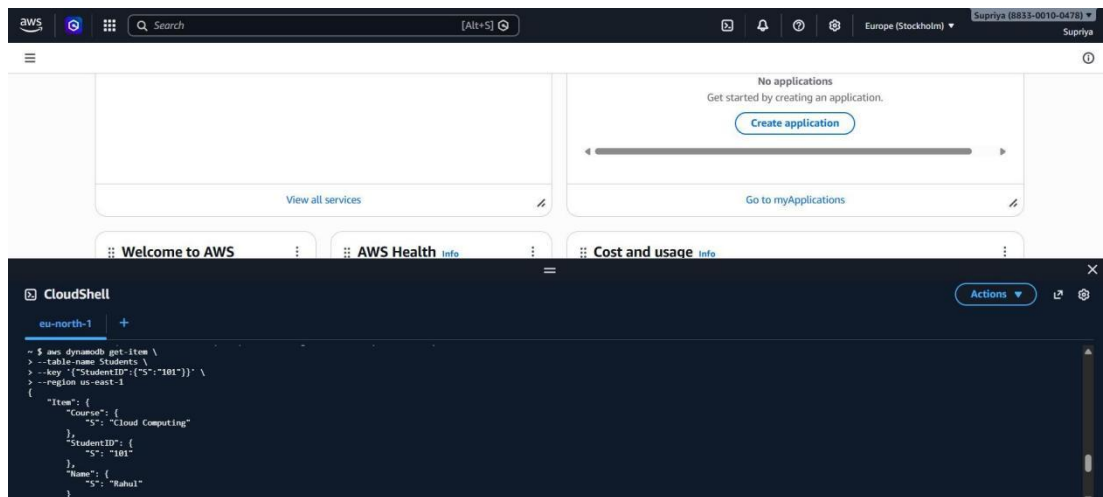
Execute the command:

aws dynamodb put-item \

--table-name Students \

--item '{"StudentID":{"S":"s101"},"Name":{"S":"Rahul"},"Course":{"S":"Cloud Computing"}}' \

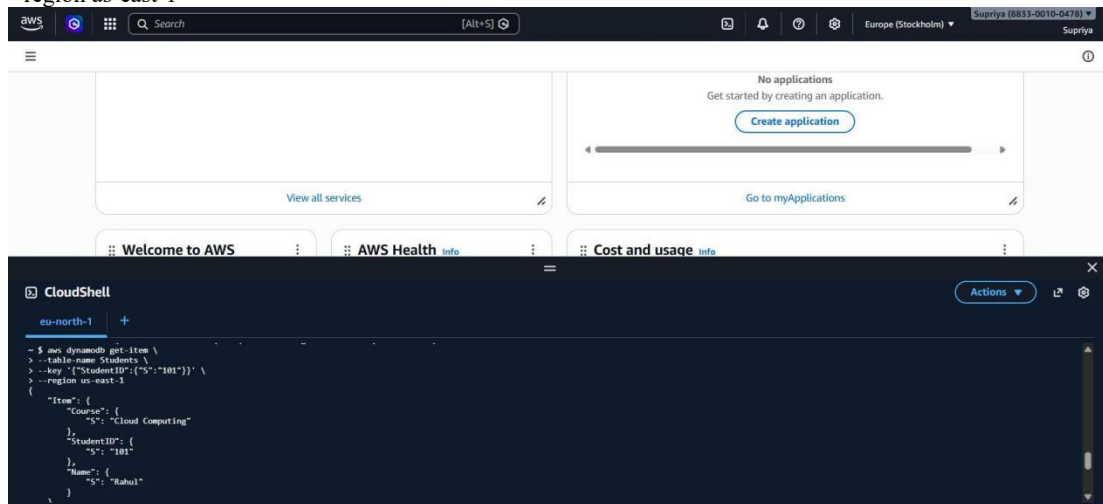
--region us-east-1



4. Retrieve Data from Table

To read the data:

```
aws dynamodb get-item \
--table-name Students \
--key '{"StudentID":{"S":"101"}}' \
--region us-east-1
```



5. Scan All Records

If you want to see **all items in the table**:

```
aws dynamodb scan --table-name Students --region us-east-1
```

6. Delete Table

```
aws dynamodb delete-table --table-name Students --region us-east-1
```

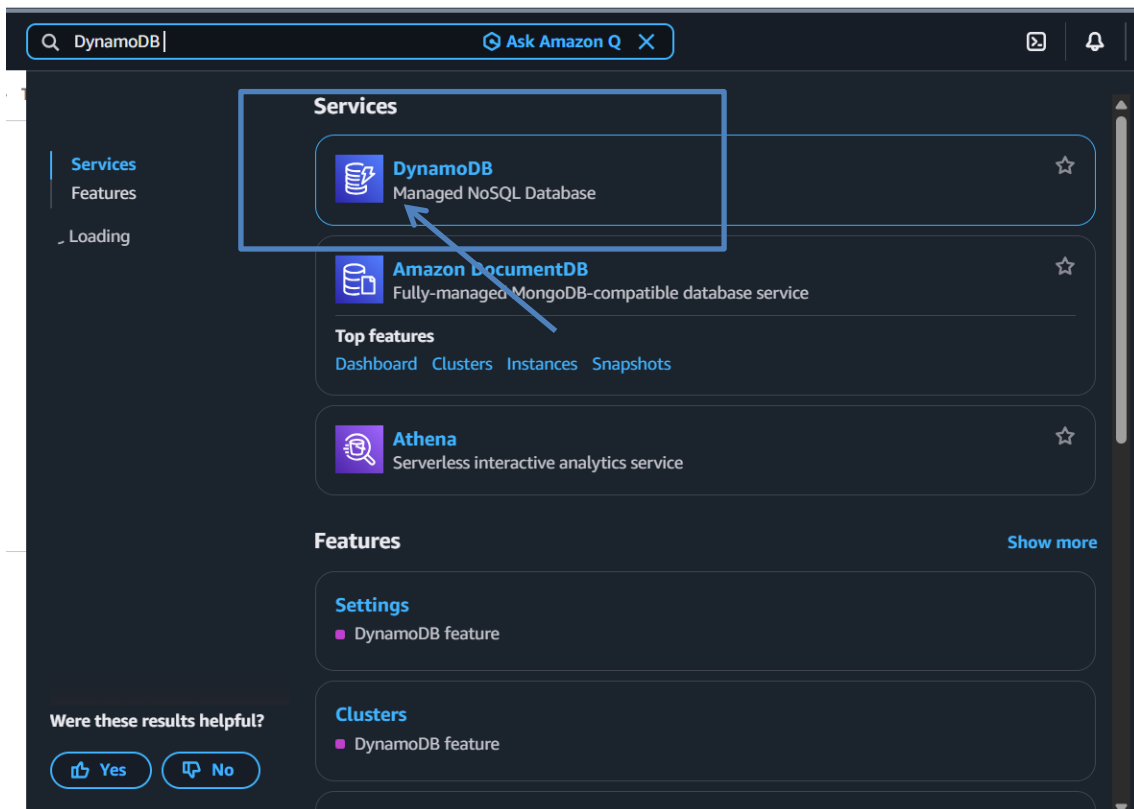
Step 1: Login to AWS Console

Open browser

Go to AWS Management Console

Login with credentials

Search DynamoDB in search bar



Step 2: Create a Table

Click Create Table

Enter:

Table Name: **Student**

Partition Key: **StudentID** (String)

Choose:

Capacity Mode → On-demand

Click Create Table

Wait until status shows Active

Table successfully created.

Rate your experience with this DynamoDB console. ☆ ☆ ☆ ☆ ☆

aws Search [Alt+S]

DynamoDB > Tables > Create table

Share your feedback on Amazon DynamoDB. Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Create table

Table details info
DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name
This will be used to identify your table.

Partition key
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

Sort key - optional
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

Rate your experience with this DynamoDB console. ☆ ☆ ☆ ☆ ☆

aws Search [Alt+S]

DynamoDB > Tables > Create table

Maximum write capacity units	-	Yes
Local secondary indexes	-	No
Global secondary indexes	-	Yes
Encryption key management	AWS owned key	Yes
Deletion protection	Off	Yes
Resource-based policy	Not active	Yes

Tags info
Tags are pairs of keys and optional values, that you can assign to AWS resources. You can use tags to control access to your resources or track your AWS spending.
No tags are associated with the resource.

[Add new tag](#)
You can add 50 more tags.

[Cancel](#) [Create table](#)

Step 3: Insert Data (Create Item)

Open **Student** table

Click Explore table items

Click Create item

Switch to **JSON View**

aws Search [Alt+S]

DynamoDB > Tables

The Student table was created successfully.

Tables (1) info
Find tables Filter by tag Filter by tag value

March 8, 2026, 22:17 (UTC+5:30) Last updated

☐	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mo
☐	Student	Active	StudentID (S)	-	0	0	Off	☆	On-demand

DynamoDB

- Dashboard
- Tables
- Explore items
- PartiQL editor
- Backups
- Exports to S3
- Imports from S3
- Integrations
- Reserved capacity
- Settings

DAX

- Clusters
- Subnet groups
- Parameter groups
- Events

Enter:

```
{  
  "StudentID": {"S": "101"},  
  "Name": {"S": "Supriya"},  
  "Department": {"S": "CSE"},  
  "Year": {"N": "3"}  
}
```

Item inserted successfully.

Step 4: Retrieve Data (Read Operation)

Click Explore table items

Use Scan option

Click Run

Stored data will be displayed.

Step 5: Update Item

Select the item

Click Edit

Change **Year** from 3 to 4

Click Save changes

Item updated successfully.

Step 6: Delete Item

Select the item

Click Delete

Confirm delete

Item removed from table

Step 7: Delete Table (Optional Cleanup)

Go back to Tables

Select **Student**

Click Delete

Type confirmation name

Confirm

Table deleted successfully.